



Visvesvaraya Technological University
BELAGAVI, KARNATAKA.

ವಿಶ್ವೇಶ್ವರಯ್ಯ ತಾಂತ್ರಿಕ ವಿಶ್ವವಿದ್ಯಾಲಯ
ಬೆಳಗಾವಿ, ಕರ್ನಾಟಕ

A PROJECT PHASE-2 REPORT

ON

“Car Parking Slot Detection Using YOLO Algorithm”

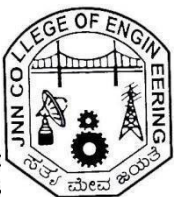
Submitted to Visvesvaraya Technological University in partial fulfillment of the requirement for the award of Bachelor of Engineering degree in Computer Science and Engineering.

Submitted by

DEEPTHI B A	4JN20CS026
DHANYATHA B	4JN20CS027
TANMAYEE SHARVANI	4JN20CS111
VARSHITHA H G	4JN20CS116

Under the guidance of

Mr. Mohan H G B.E., M.Tech.
Assistant Professor, Dept. of CS&E,
JNNCE, Shivamogga.



Department of Computer Science & Engineering
Jawaharlal Nehru New College of Engineering

Shivamogga - 577 204

May 2024

National Education Society ®



Jawaharlal Nehru New College of Engineering

Department of Computer Science & Engineering

CERTIFICATE

This is to certify that the project entitled

“Car Parking Slot Detection Using YOLO Algorithm”

Submitted by

DEEPTHI B A	4JN20CS026
DHANYATHA B	4JN20CS027
TANMAYEE SHARVANI	4JN20CS111
VARSHITHA H G	4JN20CS116

Students of 8th semester B.E. CS&E, in partial fulfillment of the requirement for the award of degree of Bachelor of Engineering in Computer Science and Engineering of Visvesvaraya Technological University, Belagavi during the year 2023-24.

Signature of Guide

Signature of HOD

Mr. Mohan H G B.E., M.Tech.

Assistant Professor, Dept. of CS&E

Dr. Jalesh Kumar M. Tech, Ph.D.

Professor and Head, Dept. of CS&E

Signature of Principal

Dr. Y. Vijaya Kumar B.E., M.Tech., Ph.D.

Principal, JNNCE

Examiners: 1. _____ 2. _____



ABSTRACT

Finding free slots in the parking area has become a major problem that the modern society is facing nowadays, which also leads to traffic overcrowding at the parking area. As the number of vehicles are increasing at a quick rate find parking slots has become difficult. In traditional parking system, a person has to find vacant slot or a gate keeper needs to inform about the available parking slot at parking area. To address this issue, a system is designed, where the user will be able to know the number of vacant slots in a particular parking area along with total number of slots at the entrance of the parking area. Today's most popular object detection model YOLO, is trained to achieve desired results with good precision scores at real-time speed. The model will be getting all the parking slots that are present in the parking area, to which marking boxes are drawn around for training. Real time processing is performed on the obtained data by the model to find whether the slot is vacant or occupied. Thus, it gives the information of available vacant slots to park the car at parking area.

ACKNOWLEDGEMENT

On presenting the Machine Learning with Application Development Major Project report on “**Car Parking Slot Detection Using YOLO Algorithm**”, we feel great to express our humble feelings of thanks to all those who have helped us directly or indirectly in the successful completion of the project work.

We would like to thank our respected guide **Mr. Mohan H.G**, Assistant Professor, Dept. of CSE, JNNCE, who have helped us a lot in completing this task, for their continuous encouragement and guidance throughout the project work.

We would like to thank our respected project coordinators **Dr. Poornima K.M** Professor, **Mr. Hiriyantha G.S**, **Mrs. Pushpa R.N** and **Dr. Ganavi M**, Assistant Professors, Dept. of CSE, JNNCE who have extended their warm support with respect to all aspects of our project.

We would like to thank **Dr. Jalesh Kumar**, Professor and Head, Dept. of CSE, JNNCE, Shivamogga and **Dr. Y. Vijaya Kumar**, Principal, JNNCE, Shivamogga for all their support and encouragement. We are grateful to the Department of Computer Science and Engineering and our institution Jawaharlal Nehru New College of Engineering and for imparting us the knowledge with which we can do our best. Finally, we also would like to thank the whole teaching and non-teaching staff of Computer Science and Engineering Department.

Thanking you all,

Project Associates,

DEEPTHI B A	4JN20CS026
DHANYATHA B	4JN20CS027
TANMAYEE SHARVANI	4JN20CS111
VARSHITHA B	4JN20CS116

TABEL OF CONTENTS

ABSTRACT		i
ACKNOWLEDGEMENT		ii
TABLE OF CONTENTS		iii
LIST OF TABLES		v
LIST OF FIGURES		vi
Chapter 1	Introduction	1-18
1.1	Overview	1
1.2	YOLO	2
1.3	Literature Survey	7
1.4	Problem Statement	17
1.5	Objectives	18
1.6	Scope	18
1.7	Organization of Report	18
Chapter 2	Domain Specification	19-21
2.1	Existing System	19
	2.1.1 Drawbacks	19
2.2	Proposed System	19
	2.2.1 Advantages	20
2.3	Requirements Specification	20
	2.3.1 Functional Requirements	20
	2.3.2 Non-Functional Requirements	21
	2.3.3 Hardware Requirements	21
	2.3.4 Software Requirements	21
2.4	Summary	21

Chapter 3	System Design and Implementation	22-28
3.1	System Design	22
	3.1.1 Design Overview	24
3.2	Implementation	25
	3.2.1 Main Packages	25
	3.2.2 Main User Defined Functions	25
	3.2.3 Algorithms of Main Module	27
3.3	Summary	28
Chapter 4	Result and Analysis	29-30
4.1	Snapshots	29
4.2	Summary	30
Chapter 5	Conclusion and Future Scope	31
5.1	Future Scope	31
Publication		32
References		33-34

LIST OF TABLES

Table. No.	Caption	Page No.
3.1	Packages	25
4.1	Output analysis	30

LIST OF FIGURES

Fig. No.	Caption	Page No.
3.1	System modules	22
3.2	System Design	24
3.3	YOLO algorithm architecture	27
4.1	Parking area image	29
4.2	Parking area with marked slots	29
4.3	Output Live video	30